SPARK MAX Specifications

The following tables provide the operating and mechanical specifications for the SPARK MAX Motor Controller.

DO NOT exceed the maximum electrical specifications. Doing so will cause permanent damage to the SPARK MAX and will void the warranty.

Main Electrical Specifications

Parameter	Min	Typ	Max	Units
Operating Voltage Range	5.5	12	24	V
Absolute Maximum Supply Voltage	-	-	30	V
Continuous Output Current	-	-	60*	A
Maximum Output Current (2 second surge)	-	-	100	A
Output Frequency	-	20	-	kHz

*Continuous operation at 60A may produce high temperatures on the heat sink. Caution should be taken when handling the SPARK MAX if it has been running at higher current level for an extended period of time.

If using a battery to power SPARK MAX, make sure the fully charged voltage is below 24V allowing for sustained operation. Some battery chemistries and configurations, including 6S LiPo packs, have a charge voltage above the maximum operating voltage for SPARK MAX.

PWM Input Specifications

Parameter	Min	Typ	Max	Units
Full-reverse Input Pulse †	-	1000	-	µs
Neutral Input Pulse ††	-	1500	-	µs
Full-forward Input Pulse †††	-	2000	-	µs
Valid Input Pulse Range	500	-	2500	µs
Input Frequency	50	-	200	Hz
Input Timeout ‡	-	50	-	ms
Default Input Deadband ‡‡	-	5	-	%
Input High Level	0.5	0.7	0.9	V
Input Voltage Max	12	-	-	V

†	Brushed: between A and B outputs at 100% duty. Brushless: A->B->C direction at 100% duty.
††	Neutral corresponds to zero output voltage (0 V) and is either braking or coasting depending on the current idle behavior mode.
†††	Brushed: between A and B outputs at 100% duty. Brushless: C->B->A direction at 100% duty.
‡	If a valid pulse isn't received within the timeout period, the SPARK MAX will disable its output.
‡‡	Input deadband is added to each side of the neutral pulse width. Within the deadband, output state is neutral. The deadband value is configurable using the REV Hardware Client or through the CAN interface.

Data Port Specifications

Parameter	Min	Typ	Max	Units
Digital input voltage range †	0	-	5	V
Digital input-high voltage †	1.85	-	-	V
Digital input-low voltage †	-	-	1.36	V
Analog input voltage range ††	0	-	3.3	V
Analog input (12bit)	-	81	-	µV
5V supply current (I5V) ‡	-	-	100	mA
3.3V supply current (I3.3V)	-	-	30	mA
Total supply current (I5V + I3.3V)	-	-	100	mA

†	See the Data Port documentation for more details on the digital pins on the Data Port.
††	See the Analog Input documentation for more details on the Data Port's analog pin.
‡	The 5V supply is shared between the Data Port and Encoder Port.

Encoder Port Specifications

Parameter	Min	Typ	Max	Units
Digital input voltage range †	0	-	5	V
Digital input-high voltage †	1.85	-	-	V
Digital input-low voltage †	-	-	1.36	V
Analog input voltage range ††	0	-	3.3	V
5V supply current (I5V) ‡	-	-	100	mA
3.3V supply current (I3.3V)	-	-	30	mA
Total supply current (I5V + I3.3V)	-	-	100	mA

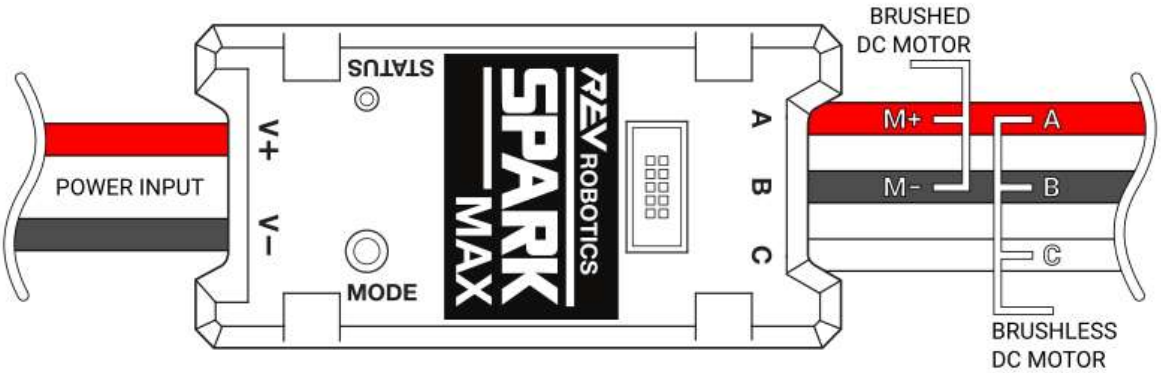
†	See the Data Port documentation for more details on the digital pins on the Data Port.
††	See the Analog Input documentation for more details on the Data Port's analog pin.
‡	The 5V supply is shared between the Data Port and Encoder Port.

Mechanical Specifications

Parameter	Min	Typ	Max	Units
Body length	-	70	-	mm
Body width	-	35	-	mm
Body height	-	25.5	-	mm
Weight	-	113.3	-	g
Power and motor wire gauge	-	12	-	AWG
Power and motor wire length	-	15	-	cm

Power and Motor Connections

SPARK MAX is designed to drive 12V brushed and brushless DC motors at currents up to 60A continuously. Power and motor connections are made through the two sets of wires built into the SPARK MAX. The wires are 12AWG ultra-flexible silicone-coated wire. Each wire runs approximately 15cm from the end faces of the controller. Be sure to take care when cutting and stripping the wires as not to cut them too short. The figure below shows these connections in detail.



SPARK MAX Motor Controller Power Connections

> As with any electrical component, make all connections with the power turned off. Connecting the SPARK MAX to a powered system may result in unexpected behavior an may pose a safety risk.

Motor Output

Motor output wires are labeled as A, B, and C with red, black, and white wires. Brushed motors must be connected to the A and B wires, while brushless motors must be connected to all three. **It is critical that the order of the brushless motor wires match the SPARK MAX or the motor will not spin and could be damaged.** Additional details are below.

Motor Connections

Motor Type	Motor Wires	SPARK MAX Wires
Brushed	Red / M+	Red / A
	Black / M-	Black / B
Brushless (NEO Brushless Motor)	Red / A	Red / A
	Black / B	Black / B
	White / C	White / C

SPARK MAX cannot detect which motor type it is connected to. Be sure to configure the SPARK MAX to run the type of motor you have connected. See the Motor Type - Brushed/Brushless Mode section for more details on configuring the appropriate motor type.

Power Input

Power input wires are labeled as V+ and V- with red and black wires. The SPARK MAX is intended to operate in a 12 V DC robot system, however, it is compatible with any DC power source between 5.5 V and 24 V.

⚠ DO NOT reverse V+ and V- or swap motor and power connections. Doing so will cause permanent damage to the SPARK MAX and will void the warranty.

⚠ DO NOT exceed the maximum supply voltage of 30V. Doing so will cause permanent damage to the SPARK MAX and will void the warranty.

When using high-current motors, it is recommended to use a power source that is capable of handling large surge currents, e.g. a 12V lead-acid battery. If the supply voltage drops below 5.5V the SPARK MAX will brown out, resulting in unexpected behavior. It is also highly recommended to incorporate a fuse or circuit breaker in series with the SPARK MAX between it and the power source to prevent exceeding the maximum current rating.

DO NOT exceed the maximum current ratings:

- 60A for 3 minutes
- 100A for 2 seconds

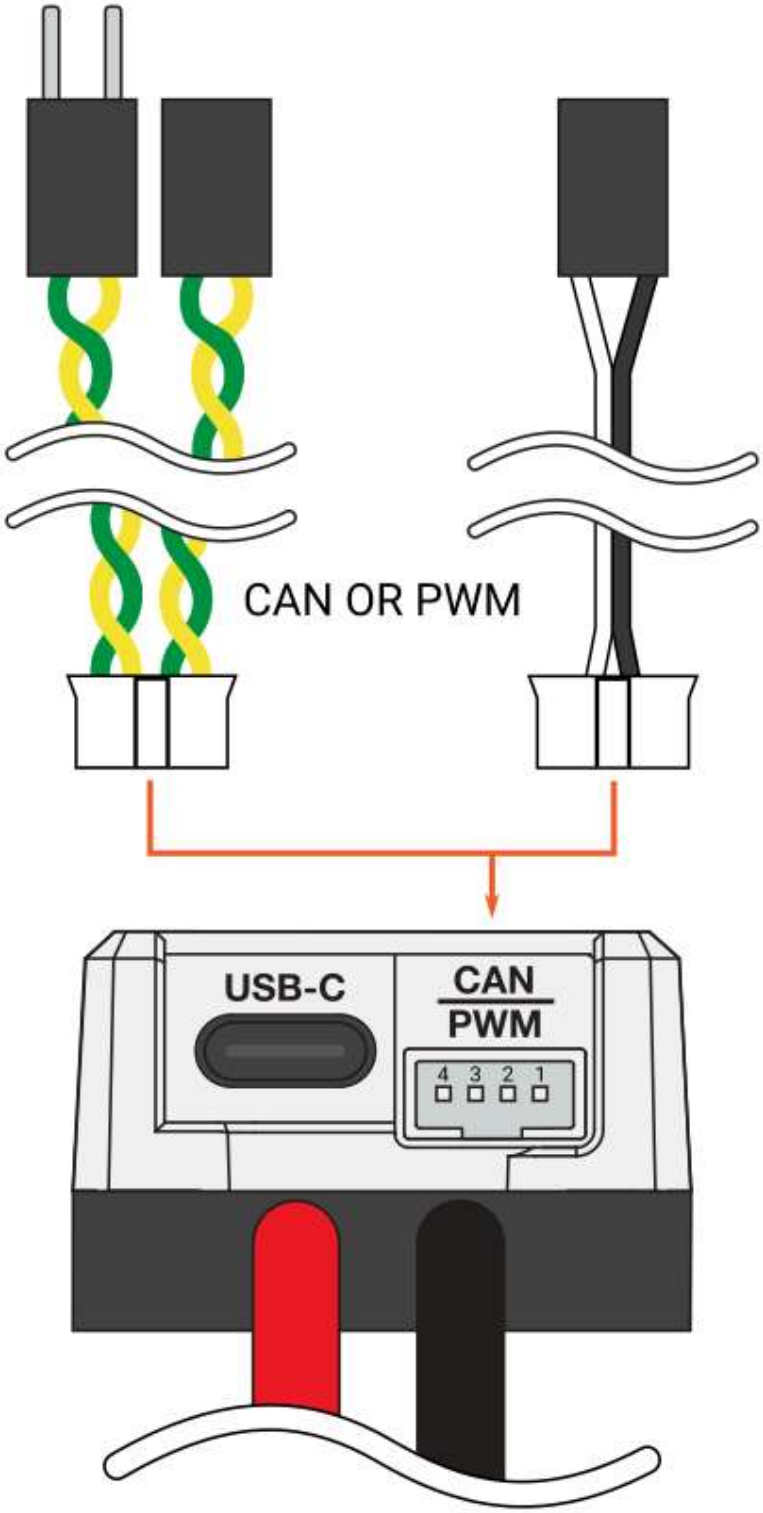
Doing so will cause permanent damage to the SPARK Flex and will void the warranty.

Control Connections

The SPARK MAX can be controlled by three different interfaces, servo-style PWM, controller area network (CAN), and USB. The following sections describe the physical connections to these interfaces in detail. For details on the operation and protocols of the PWM, CAN, and USB interfaces, please see the section on Control Interfaces.

CAN/PWM Port

The CAN/PWM Port is located on the power input side of the SPARK MAX. This port can be connected to either a servo-style PWM signal or a CAN bus with other devices. Connector details can be found below.


CAN OR PWM
USB-C
CAN
PWM
4 3 2 1

CAN/PWM Port Connector Information

Connector Pin	CAN Function	PWM Function
1	CAN High	Signal
2	CAN Low	Ground
3	CAN High	Signal
4	CAN Low	Ground

Mating Connector Information

Description	Manufacturer	Part Number	Vendor	Vendor P/N
JST-PH 4-pin Housing	JST	PHR-4	DigiKey	455-1164-ND ↗
JST-PH Contact	JST	SPH-002T-P0.5L	DigiKey	455-2148-1-ND ↗
Recommended Crimping Tool	IWISS	SN-2549	Amazon	SN-2549 ↗

Identical-function pins are electrically connected inside the SPARK MAX, therefore the CAN daisy-chain is completed internally and any two signal and ground pairs can be used for PWM.

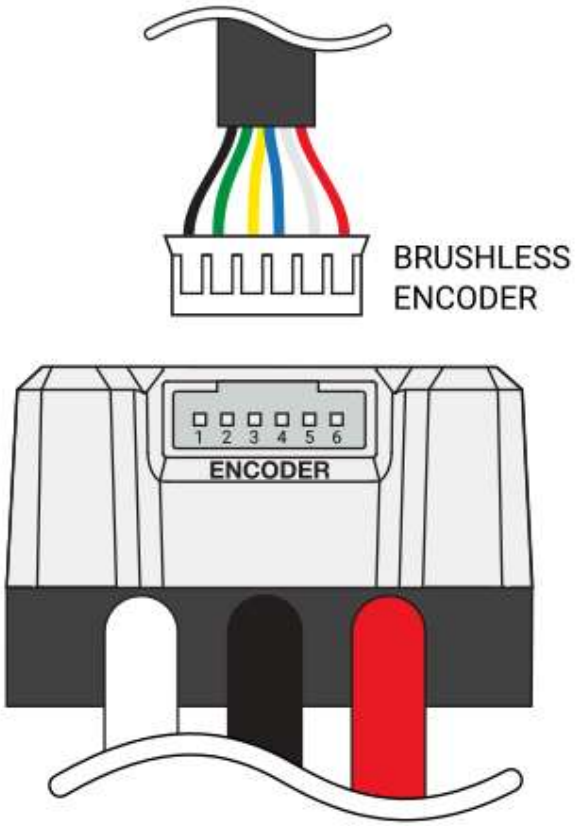
USB-C Port

The USB-C Port is located on the power input side of the SPARK MAX. It supports USB 2.0 and 5V power for the SPARK MAX's internal microcontroller. While you can configure the SPARK MAX without main power, you will not be able to spin a motor.

Encoder Port

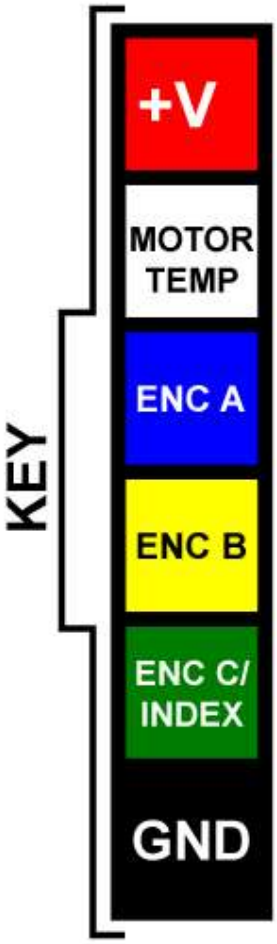
Located on the motor output side of the SPARK MAX is a 6-pin Encoder Port. This port is designed to accept the built-in hall-encoder from the NEO Brushless Motor ↗, but it can also connect to other external encoders when running in Brushed Mode. The connector details can be found below.

> The SPARK MAX can be configured to run in Alternate Encoder Mode, which reconfigures the Data Port on the top of the controller to accept an alternative quadrature encoder in addition to the Encoder Port.



Encoder Port Connector Information

Connector Pin	Pin Type	Pin Function
1	Power	Ground
2	Digital	Encoder C / Index
3	Digital	Encoder B
4	Digital	Encoder A
5	Analog	Motor Temperature
6	Power	+5V



ENCODER PORT PINOUT WITH JST PH 6-PIN EXTENSION CABLE (REV-11-1275-PK4)

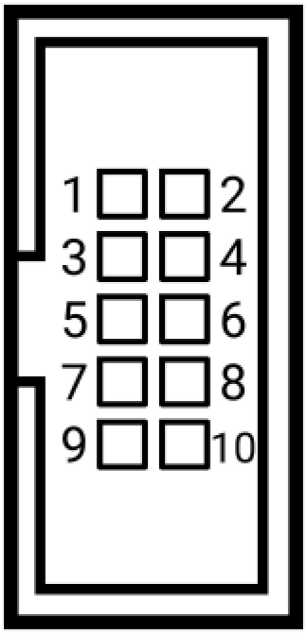
Mating Connector Information

Description	Manufacturer	Part Number	Vendor	Vendor P/N
JST-PH 6-pin Housing	JST	PHR-6	DigiKey	455-1162-ND ↗
JST-PH Contact	JST	SPH-002T-P0.5L	DigiKey	455-2148-1-ND ↗
Recommended Crimping Tool	IWISS	SN-2549	Amazon	SN-2549 ↗

Data Port

SPARK MAX Data Port Pinout

Located on the top of the SPARK MAX, the Data Port allows for extra sensor input and future feature development. The connector details can be found below.



Connector Pin	Pin Type	Pin Function
1	Power	+3.3V
2	Power	+5V
3	Analog	Analog Input
4	Digital	Forward Limit Switch Input
5	Digital	Encoder B
6	Digital	Multi-function Pin
7	Digital	Encoder A
8	Digital	Reverse Limit Switch Input
9	Digital	Encoder C / Index
10	Ground	Ground

Using the SPARK MAX Data Port Breakout Board (REV-11-1278 ↗) makes interfacing with the SPARK MAX Data Port easier.

SPARK MAX Data Port Features

Limit Switch Inputs

SPARK MAX has two limit switch inputs that, when triggered, can independently prevent motion in both the forward and reverse directions. By default, when the pin for the corresponding direction is grounded, SPARK MAX will override any input commands for that direction and force the output into the neutral state. Input commands for the opposite direction will still be processed unless the corresponding limit signal is also triggered.

The default polarity is compatible with Normally Open (NO) style limit switches, whose contacts are shorted together when the switch is pressed. The Limit Switch Inputs can be configured for the opposite polarity using the USB or CAN interfaces. When configured for the opposite polarity, Normally Closed (NC), the limit will be triggered when the pin is left disconnected from ground. In other words, connecting the pin to ground will release the limit. The following table shows these configurations in detail:

Limit Switch Operation

Connected Switch Type	Configured Limit Switch Polarity	Switch State	Limit Triggered (preventing motion)	Recommended Combinations
Normally Open	Normally Open (default)	Released	Normal operation	*
		Pressed	Triggered	
	Normally Closed	Released	Triggered	
		Pressed	Normal operation	
Normally Closed	Normally Open (default)	Released	Triggered	
		Pressed	Normal operation	
	Normally Closed	Released	Normal operation	*
		Pressed	Triggered	

Quadrature Encoder Input

The Quadrature Encoder Input on the Data Port is compatible with standard quadrature encoder signals, usually labeled as channel A, channel B, and Index. SPARK MAX shares these signals with the Encoder Port on the output side of the controller, therefore the Index signal is shared with the third brushless encoder signal C. When in Brushless Mode, these Data Port pins cannot be used with an external encoder. See Alternate Encoder Mode for information on how to configure the SPARK MAX to accept an alternative encoder source when running in Brushless Mode.

When in Brushed Mode, an external encoder can be connected through either the Data Port or the Encoder Port.

The SPARK MAX encoder signals are not pulled high internally. This is to ensure the maximum compatibility with different types of encoders.

Analog Input

The Analog Port on the SPARK MAX can measure voltages up to 3.3V with 12-bit resolution. The SPARK MAX Data Port Breakout includes a 5V to 3.3V amplifier circuit so that 5V signals can be sensed with the Analog Input pin.

Analog input is supported on firmware versions 1.4.0 and newer.

Multi-function Pin

This pin is reconfigured when the SPARK MAX is configured in Alternate Encoder Mode.

Power Rails

The SPARK MAX Data Port can provide both 3.3V and 5V power to connected devices. Please check Data Port Specifications for details on the supply current capabilities of both rails.

SPARK MAX Data Port Accessories

Alternate Encoder Adapter

Features

The SPARK MAX Alternate Encoder Adapter (REV-11-1881-PK2 ↗) enables the use of an alternative encoder source different from the default. This is especially useful when running one of the NEO Brushless Motors, as the default encoder port is occupied by the built-in NEO hall sensors. Please see the Alternate Encoder Mode section in the SPARK MAX User's Manual for more information.

- JST PH 6-pin connector

- Pinout compatible with REV Through Bore Encoder
- Index Signal/Absolute PWM Pulse selection switch
- Selects which signal is connected to pin 4 of the Data Port
- Solder pads
- Analog Input
- 3.3V and 5.0V Power
- Ground

Specifications

- 1 x JST PH, 6-pin connector
- 1 x 14 Position 2 Row Receptacle Connector 0.050"

Absolute Encoder Adapter

Features

The SPARK MAX Absolute Encoder Adapter (REV-11-3326 ↗) connects the Absolute Duty Cycle output of the Through Bore Encoder to the correct SPARK MAX Data Port pins, leaving the incremental quadrature pins disconnected.

- JST PH 6-pin connector
- Pinout Compatible with REV Through Bore Encoder
- Solder pads
- Limit Switches
- Ground

Specifications

- 1 x JST PH, 6-pin connector
- 1 x 14 Position 2 Row Receptacle Connector 0.050"

Data Port Breakout Board

Features

The SPARK MAX Data Port Breakout Board (REV-11-1278↗) makes it easy to connect external sensors to the SPARK MAX Data Port.

- Solder pads for every Data Port pin
- Analog input 5V to 3.3V converter
  - Built-in amplifier maps 0V - 5V analog signals to the native 0V - 3.3V range of the SPARK MAX Analog Input
  - Configurable resistors can bypass the amplifier (move R3 to R4 position)
- Pass-through Data Port connector
  - Connect other sensors with data port-compatible cables while using this breakout
- Mounts directly to SPARK MAX
  - No need for a data port cable
  - Securely mounts to the SPARK MAX zip-tie notches